

CYTOTOXICITIES OF METABOLITES FROM A *MONO CILLIUM* SPECIES

PROMA KHONDKAR, MOHAMMAD MUKHLESUR RAHMAN
AND MOHAMMAD ANWARUL ISLAM*

Department of Pharmacy, University of Rajshahi
Rajshahi-6205, Bangladesh.

ABSTRACT

The cultural broth of a *Monocillium* species upon extraction with ethyl acetate afforded an antimicrobial compound, 5-hydroxymethyl furfural (1). Both the ethylacetate extract and isolated compound (1) showed strong cytotoxicities in brine shrimp lethality bioassay. The LC_{50} values of ethyl acetate extract and compound 1 were determined graphically and found to be 14.16 $\mu\text{g/ml}$ and 27.99 $\mu\text{g/ml}$, respectively.

INTRODUCTION

To combat infectious diseases due to pathogenic microorganisms, medicine today makes the use of antibiotics which, in turn, may be produced by microorganisms like bacteria, fungi, actinomycetes. Among the fungi, *Penicillium* and *Aspergillus* produced the highest number of antibiotics (Gennaro, 1990). However, among the members of the genera *Monocillium*, only *Monocillium nordinni* was reported to produce metabolites with antifungal property (Ayer *et al.*, 1980), hepatocarcinogenicity (Ayer *et al.*, 1981). Here, we present the cytotoxicities of ethyl acetate extract of the cultural broth of a *Monocillium* species as well as of 5 hydroxymethylfurfural isolated from this extract using brine shrimp lethality bioassay techniques (McLaughlin, 1982; McLaughlin, 1988; Meyer *et al.*, 1982).

Brine shrimp bioassay is a recent development for assessment of cytotoxicity of bioactive compounds, as well as extracts. Thus, *in vitro* mortality on a zoologic organism, *Artemia salina*, may be used as a convenient monitor for the screening and fractionation for the development of new bioactive natural products.

MATERIALS AND METHODS

Isolation of the Organism: After screening a number of soil samples of different depths and places of Bangladesh, an antagonist organism was isolated which was identified as a *Monocillium* species on the basis of its morphological characteristics (Anisworth and Bisley, 1993).

Extraction, Isolation and Identification of Compound I:

The cultural broth of *Monocillium* species after 20 days at 37°C was extracted with ethyl acetate. This extract was concentrated *in vacuo* at 45°C to get semisolid mass. The thin layer chromatography of this extract with a solvent system of ethyl acetate and Petroleum ether (7:13) showed three distinct compounds of which only one exhibited antimicrobial activity. This compound was isolated by preparative thin layer chromatography with the same solvent (Khondkar, 1988).

The compound was identified as 5-hydroxymethylfurfural by comparing its UV, IR, H-NMR and ^{13}C -NMR data to those of an authentic compound which was previously isolated from *Talaromyces flavus* (Ayer and Racok, 1990).

Cytotoxicity Screening:

The eggs of the brine shrimp, *Anemia*

*Correspondence

sauna, were collected from an aquarium shop (Katabon; Dhaka, Bangladesh) and hatched in the Department of Pharmacy, Rajshahi University, Bangladesh. To hatch, sea water (prepared by dissolving 3.8 gm NaCl in 100 ml distilled water) was placed in a small tank divided into two parts with a net. The eggs were kept in one part of the tank and a light source was applied in other part to attract the shrimp through the pores of net. After 48 hrs, the shrimps were matured as nauplii (Meyer *et al.*, 1988).

Both ethyl acetate extract and compound 1 were used as test samples for this investigation. Each of these samples was first dissolved in very small amount of DMSO. Then five concentrations, viz. 2.5 µg/ml, 5µg/ml, 10 µg/ml, 20 µg/ml, 40 µg/ml and 80 µg/ml, were obtained by adding sea water in such a way that each sample concentration attained of final volume of 5 ml. However, not more than 500 DMSO) should be present in a vial having 5 ml of final volume.

In each test vial. 20 matured shrimps were placed. For the assessment of the accuracy of the test, one vial containing same volume of DMSO plus sea water for each concentration was used as a control group. After 24 hrs. the number of survived nauplii were counted in both experimental and control groups.

RESULT AND DISCUSSION

The concentrated ethyl acetate extract of the cultural broth of *Monocillium* species was subjected to preparative thin layer chromatography to isolate compound 1 which was identified as 5-hydroxymethylfurfural.

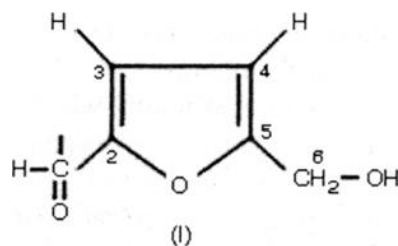


Table 1
Brine Shrimp Lethality Bioassay of Ethylacetate Extract and Compound 1

Test Sample	Conc. of Sample (µg/ml)	LogC	No. of Shrimp added	% of Mortality after 24hrs	LC ₅₀ from graph
	2.5	0.39	20	10	
	5.0	0.69	20	20	
	10	1.00	20	40	
Ethylacetate Extract	20	1.30	20	60	14.16
	40	1.60	20	80	
	80	2.00	20	100	
	2.5	0.39	20	5	
	5.0	0.69	20	10	
	10	1.00	20	20	
Compound-1	20	1.30	20	40	27.99
	40	1.60	20	60	
	80	2.00	20	80	

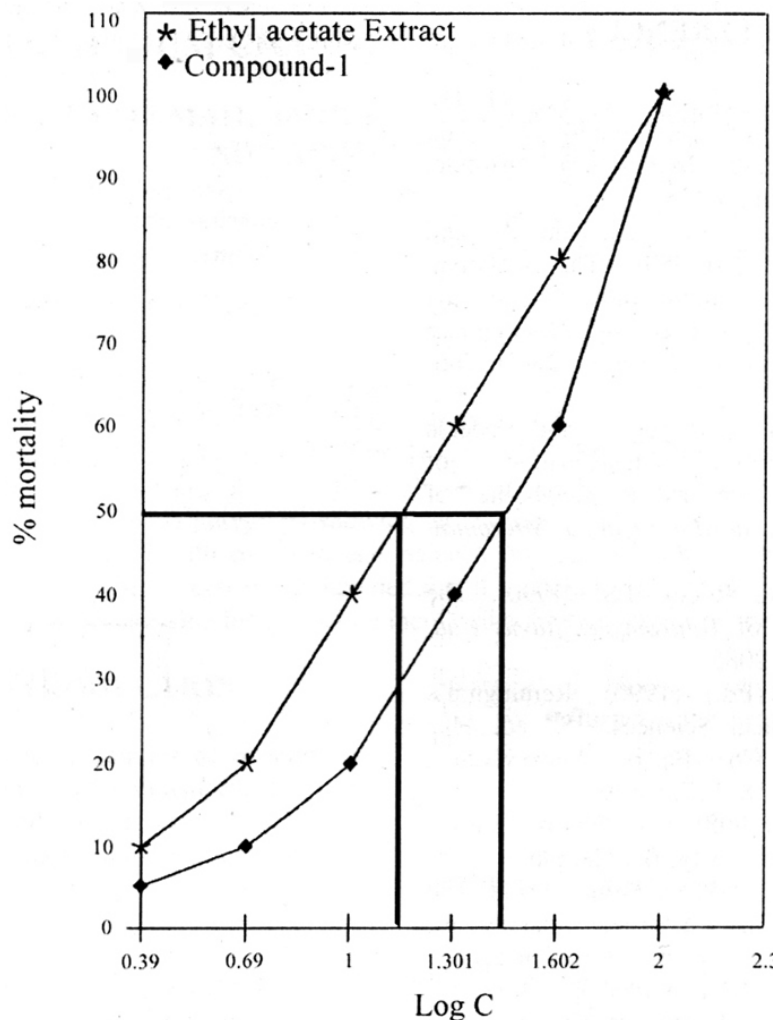


Fig. 1: Graphical representation of LC₅₀ values in Brine Shrimp Lethality bioassay.

The number of nauplii that died after 24 hrs of sample application was counted and percentage of mortality was calculated for each concentration of experimental group. In control group no mortality of nauplii was observed. The overall data for this investigation are presented in Table-1. From the data, it is evident that the rate of mortality of the nauplii was found to be increased with the increase in concentration of the test samples. A plot of log of concentration versus percentage of mortality of nauplii gave an almost linear relationship (Fig.1). From the graph, LC₅₀ values of ethyl acetate extract and 5-hydroxymethylfurfural were calculated by

extrapolation and found to be 14.16µg/ml and 27.99µg/ml respectively. From the result, it is evident that the ethyl acetate extract was more cytotoxic than the isolated compound. So, we can conclude that a more cytotoxic compound might be presents in ethyl acetate extract.

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